

Mock exam 1

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In vitro fertilization (IVF) and pregnancy tests

A girl underwent IVF, and she wants to know whether she is pregnant or not. From the previous studies, we know that the chance of conception after IVF is 0.222 (CDC, 2021).

She used an early pregnancy detection kit that gives the correct answer 95% time if the person is pregnant and 80% time if the person *is not* pregnant. That is: it detects the elevated level of hCG in 95% of cases and does not react below the threshold value in 80% of cases. Answer the questions below and provide clear calculations.

Questions

1. What is the chance that a girl with a positive test is really pregnant? Is this result convincing? (10 points.)
2. What is the chance that a girl with a *negative* test is still pregnant? (3 points.)
3. She decided to be double sure. After some time, she used the same test once again, and she got a positive result *once again*. What would be the chance that she is pregnant given she got positive results *twice*? (5 points.)
4. What can you say about this test kit? Is it good? Is it specific? Is it sensitive? Support your claims with some calculations. **Note:** use odds and odds ratios to support your words. (7 points.)

Marking

This task will help to demonstrate the use of the Bayes formula that is shown below.

$$P_{(A \mid B)} = \frac{P_{(B \mid A)} \cdot P_{(A)}}{P_{(B)}}$$

$$P_{(B)} = \sum P_{(B_i \mid A_i)} \cdot P_{(A_i)}$$

As A and B do not look clear, we can assume B in the Bayes formula as the test^+ and A as the girl being pregnant. Thus, $P_{(A \mid B)}$ represents the chance of the girl being pregnant given the pregnancy test is positive. We do not know it. But we know $P_{(B \mid A)}$, the chance of the test to be positive (test^+) if the person is pregnant (the correct test^+).

Let's summarize what we have.

We have a system with two events each having two possible outcomes: pregnancy (pregnant vs non-pregnant) and test result (test is positive or negative). Thus, the formula becomes easier to grasp:

$$P_{(A \mid B)} = \frac{P_{(B \mid A)} \cdot P_{(A)}}{P_{(B)}}$$

$$P_{(\text{pregnant} \mid \text{test}^+)} = \frac{P_{(\text{test}^+ \mid \text{pregnant})} \cdot P_{(\text{pregnant})}}{P_{(\text{test}^+)}}$$

We can build a table to visualize the information we have (Table 1):

Table 1: The data we have initially.

	Not pregnant	Pregnant	$P_{(\text{test})}$
Test –	(0.8 not pregnant)		
Test +		(0.95 0.222)	
$P_{(\text{preg})}$		0.222	1

What is the chance that a girl with a positive test is really pregnant? Is this result convincing? (10 points.)

So, we know $P_{(A)}$, the chances of pregnancy, $P_{(B | A)}$, the chance of the test to be positive (test⁺) if the person is pregnant (the correct test⁺), $P_{(B^c | A^c)}$, the chance of the test to be positive (test[–]) while the person is not pregnant (correct test[–]). Not bad. For this task, we need to use the Bayes formula. The only element we miss is $P_{(B)}$, the chances of test⁺, either if the girl is pregnant (correct test⁺) or not (*wrong* test⁺, false positive result). We need to dive into the conditional probability.

Let's do it!

The probability of being pregnant *AND* tested positive can be determined by the intersection formula: $P_{(\text{test}^+ \cap \text{preg})} = P_{(\text{test}^+ | \text{preg})} \cdot P_{(\text{preg})} = 0.2109$. To find out the probability of not being pregnant *AND* still tested positive, we need to use the formula: $P_{(\text{test}^+ \cap \text{not preg})} = P_{(\text{test}^+ | \text{not preg})} \cdot P_{(\text{not preg})}$. But we do not know this term: $P_{(\text{test}^+ | \text{not preg})}$. Luckily, we know its complement, $P_{(\text{test}^- | \text{not preg})} = 0.8$. In this condition, being *NOT* pregnant, we disregard another state (*being pregnant*) and consider the chosen condition as 100%. Thus, the complement to $P_{(\text{test}^- | \text{not preg})}$ is $P_{(\text{test}^+ | \text{not preg})} = 1 - 0.8$. Our table now looks as follows (Table 2):

Table 2: We determined several terms.

	Not pregnant	Pregnant	$P_{(\text{test})}$
Test –	(0.8 not pregnant)		
Test +	(0.2 not pregnant)	0.2109 (0.95 0.222)	
$P_{(\text{preg})}$		0.222	1

The overall chance of being not pregnant after IVF, $P_{(A^c)} = 1 - P_{(A)} = 1 - 0.222$. $P_{(\text{test}^+)} = P_{(\text{test}^+ | \text{preg})} + P_{(\text{test}^+ | \text{not preg})}$. Now, $P_{(\text{test}^+ | \text{not preg})}$ is easy to determine. Finally, our table will look as follows (Table 3):

Table 3: All elements for the first task are determined!

	Not pregnant	Pregnant	$P_{(\text{test})}$
Test –	(0.8 0.778)		
Test +	0.1556 (0.2 0.778)	0.2109 (0.95 0.222)	0.3665
$P_{(\text{preg})}$	0.778	0.222	1

Finally, the Bayes formula would look as follows:

$$P_{(pregnant | test^+)} = \frac{P_{(test^+ | pregnant)} \cdot P_{(pregnant)}}{P_{(test^+)}} = \frac{0.95 \cdot 0.222}{0.3665} \approx 0.5754$$

Is this result convincing?

Not really. Basically, it means that the girl may be pregnant or not with almost equal probability (0.5754 and 0.4246, respectively; OK, the probability of pregnancy is slightly higher). Still much better than before (0.222) that is rather low.

What is the chance that a girl with a *negative* test is still pregnant? (3 points.)

We want to find out $P_{(pregnant | test^-)}$. We need to:

1. Calculate the chances of being pregnant (already known and is 0.222) and *not* pregnant (1 - 0.222);
2. Probability of tested negative given the person *is* pregnant or *is not* pregnant;
3. The overall, marginal, probability of the negative result of the test.

We can update Table 3 further and get the following result (Table 4):

Table 4: All the table elements are added.

	Not pregnant	Pregnant	$P_{(test)}$
Test –	0.6224 (0.8 0.778)	0.0111 (0.05 0.222)	0.6335
Test +	0.1556 (0.2 0.778)	0.2109 (0.95 0.222)	0.3665
$P_{(preg)}$	0.778	0.222	1

Finally, the Bayes formula would look as follows:

$$P_{(pregnant | test^-)} = \frac{P_{(test^- | pregnant)} \cdot P_{(pregnant)}}{P_{(test^-)}} = \frac{0.05 \cdot 0.222}{0.6335} \approx 0.0175$$

Thus, if the person gets a negative test result, it is rather unlikely that they are pregnant. This posterior probability would be difficult to challenge even with several subsequent positive tests. Try it by yourself! $Test^+$ may mean almost anything, but $test^-$ means exactly what it is.

She decided to be double sure. After some time, she used the same test once again, and she got a positive result *once again*. What would be the chance that she is pregnant given she got positive results *twice*? (5 points.)

As our expectations changed (*we saw the results of the first test*), we may change our prior belief for the second test to the posterior belief from our previous step: from 0.222 to 0.5754. Chances of the correct (or wrong) result of the test do not change (as she is using the same kit), but as the overall prior belief in the pregnancy changed, we need to recalculate the whole table (Table 5):

Table 5: Previous posteriors are our new priors.

	Not pregnant	Pregnant	$P_{(\text{test})}$
Test –			
Test +	0.0849 (0.2 0.4246)	0.5466 (0.95 0.5754)	0.6315
$P_{(\text{preg})}$	0.4246	0.5754	1

So, our Bayes formula will look as follows.

$$P_{(\text{preg} | \text{test}^{++})} = \frac{P_{(\text{test}^+ | \text{preg})} \cdot P_{(\text{preg} | \text{test}^+)}}{P_{(\text{test}^+ \text{ updated})}} \approx$$

$$\approx \frac{0.95 \cdot 0.5754}{0.6315} \approx 0.8656$$

The chances that the girl is really pregnant increase with each positive test! But they would increase faster if the test was better. How? See below!

What can you say about this test kit? Is it good? Is it specific? Is it sensitive? Support your claims with some calculations. Note: use odds and odds ratios to support your words. (7 points.)

The test kit is not good. Why? The chances of the *correct results* are quite low for the positive test. How low? Let's calculate the odds of the correct result for each condition: for a positive test (test^+) and for a negative test (test^-).

$$\text{Odds}_{(\text{pregnant vs non-pregnant AND test}^+)} = \frac{0.211}{0.1556} \approx 1.356$$

$$\text{Odds}_{(\text{non-pregnant vs pregnant AND test}^-)} = \frac{0.6222}{0.0111} \approx 56.054$$

Thus, the odds of getting the correct results for the positive test are extremely low:

$$\text{Odds ratio} = \frac{1.356}{56.054} \approx 0.024$$

This test kit is too sensitive. Chances of false positive results are too high. It would be better to increase the chance of making the correct decision if the person is not pregnant == decrease the rate of false positive results.

Originally created by Dmytro Shytikov in 2024.